

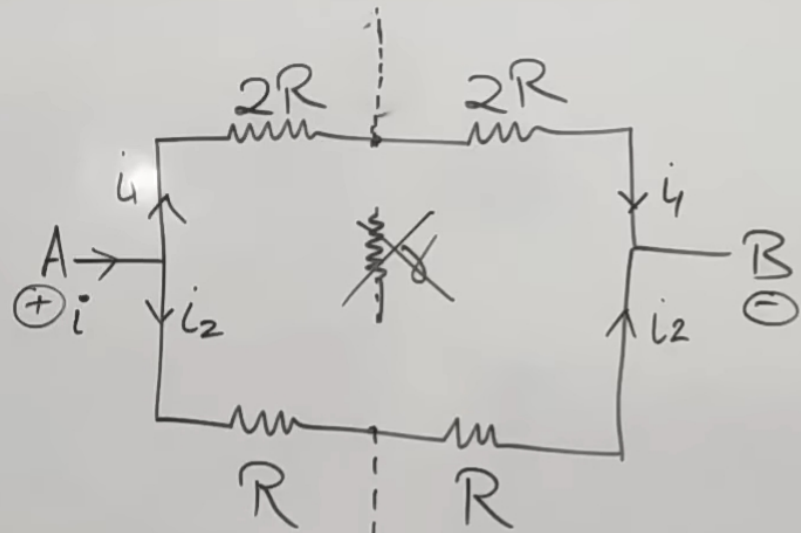
Symmetry Rules.

① Mirror Symmetry

$$R_e = \frac{R_1 R_2}{R_1 + R_2}$$

$$= \frac{4R \cdot 2R}{4R + 2R} = \frac{8R^2}{6R}$$

$$= \frac{4}{3}R$$

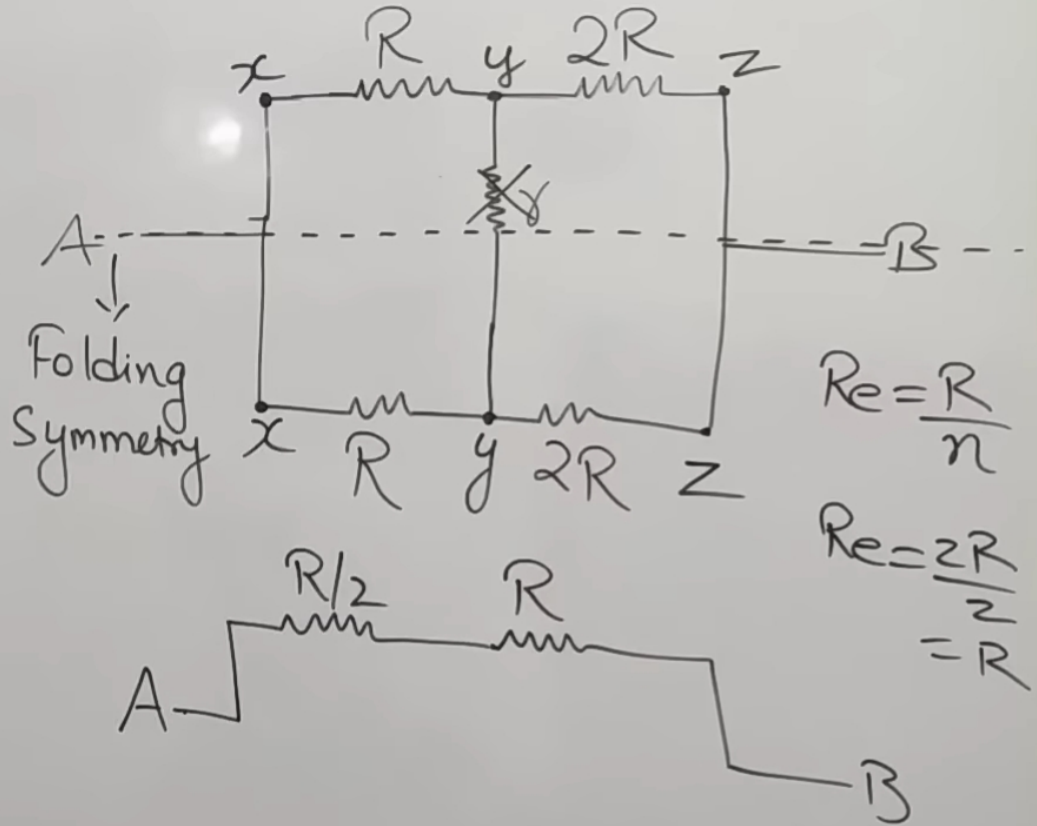


→ Mirror Symmetry
4R & 2R in parallel.

Symmetry Rules.

② Folding Symmetry

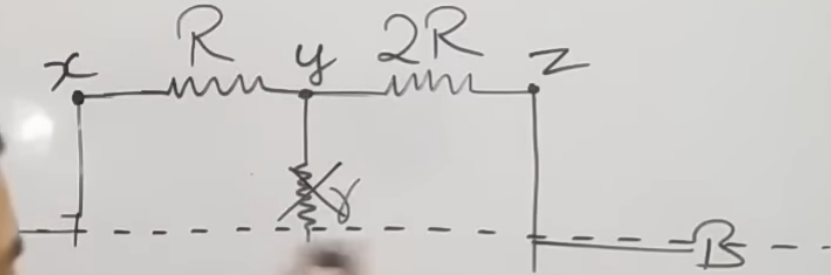
$$R_e = \frac{R}{2} + R$$
$$= \frac{3R}{2}$$



Symmetry Rules.

② Folding Symmetry

$$R_e = R$$



$$R_e = \frac{R}{n}$$

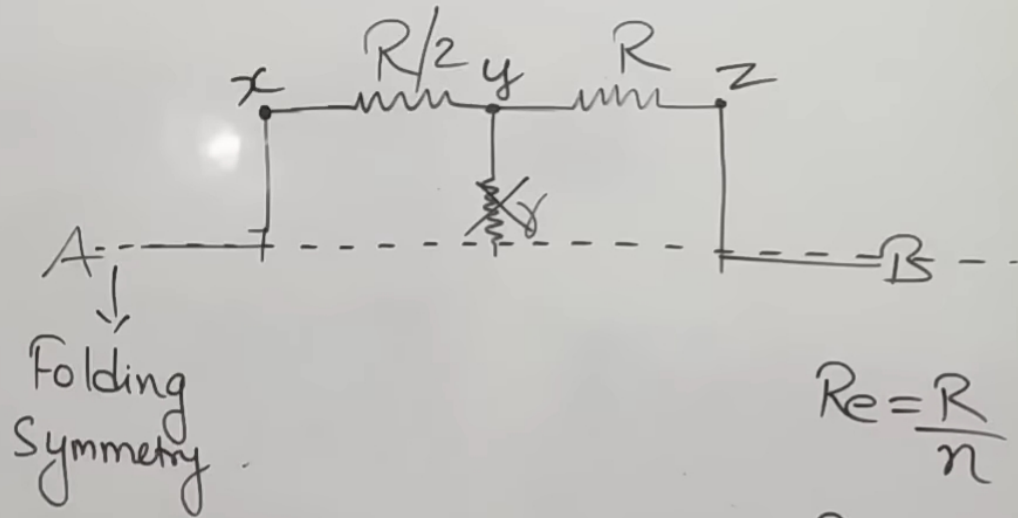
$$R_e = \frac{2R}{2} = R$$

B

Symmetry Rules.

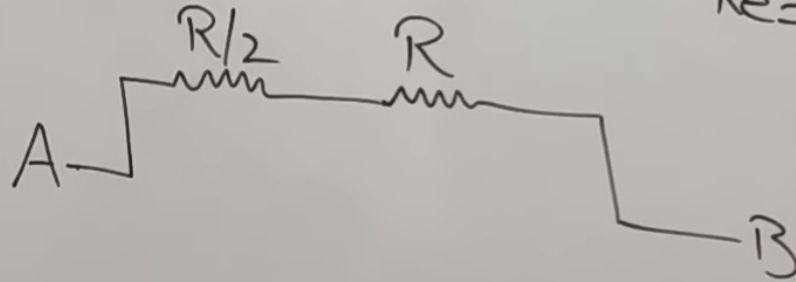
② Folding Symmetry

$$R_e = \frac{R}{2} + R$$
$$= \frac{3R}{2}$$



$$R_e = \frac{R}{n}$$

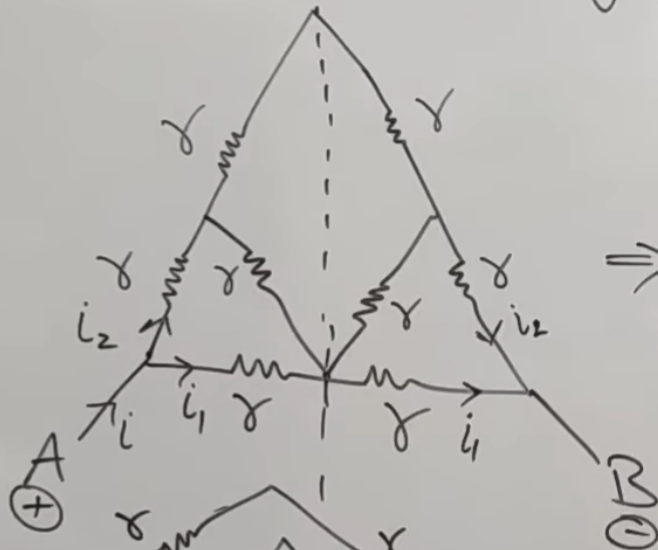
$$R_e = \frac{2R}{2}$$
$$= R$$



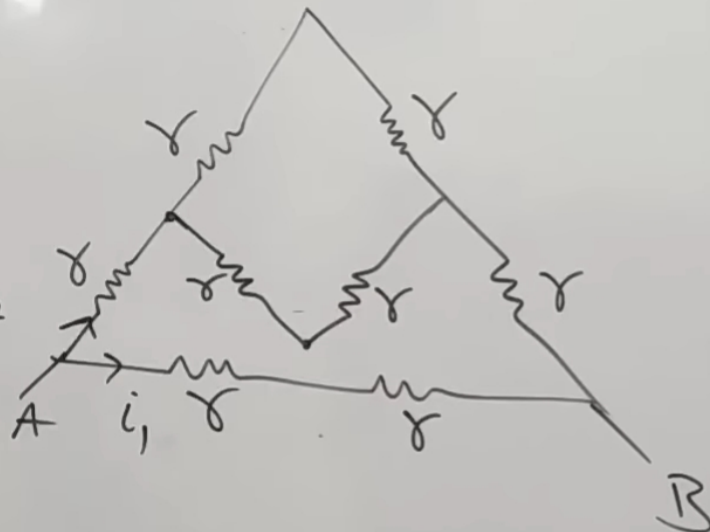
Maxwell Symmetry.

→ Maxwell symmetry

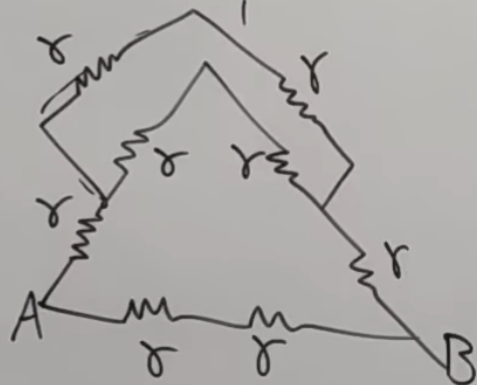
①



$\Rightarrow$



$\Rightarrow$



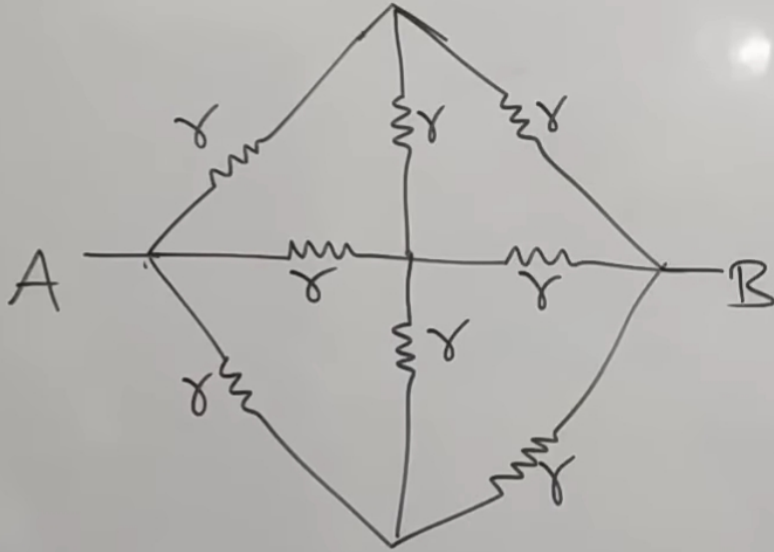
$\Rightarrow$



$$R_e = \frac{3r \cdot 2r}{3r + 2r} = \frac{6r}{5}$$

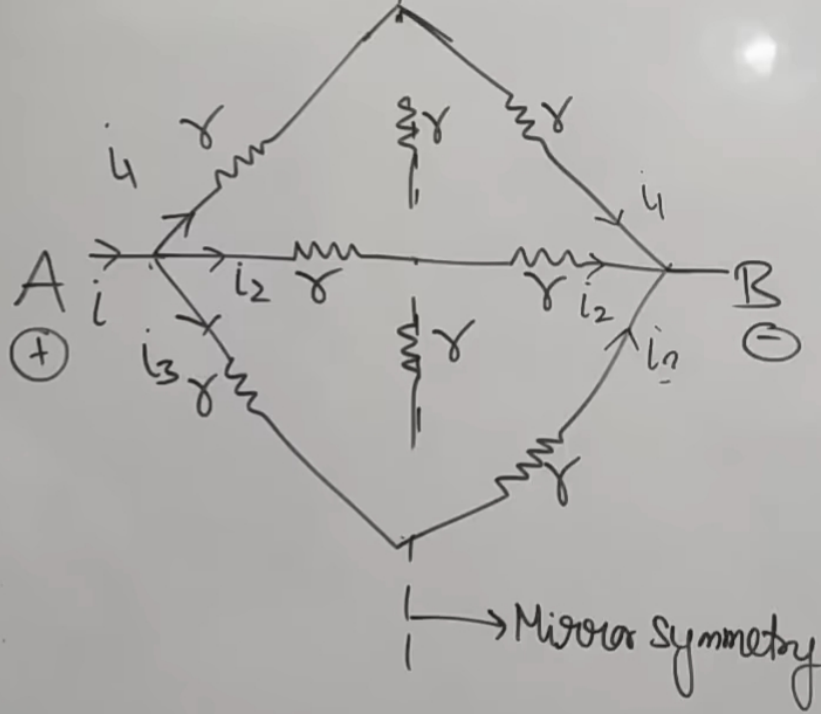
Maxwell Symmetry:

②



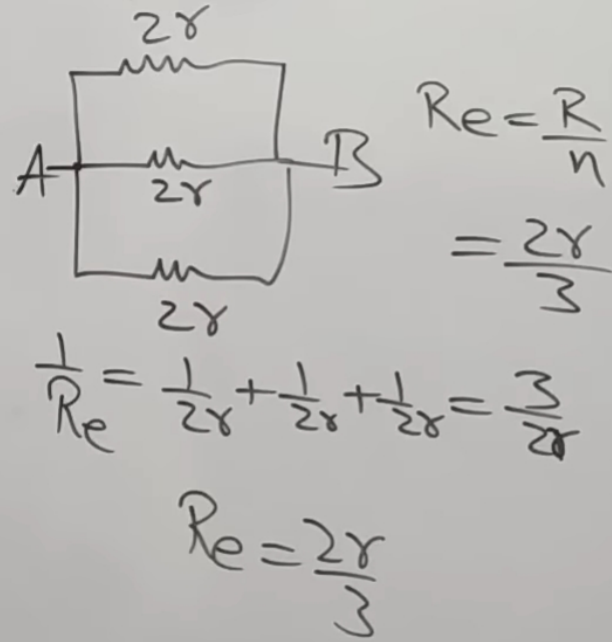
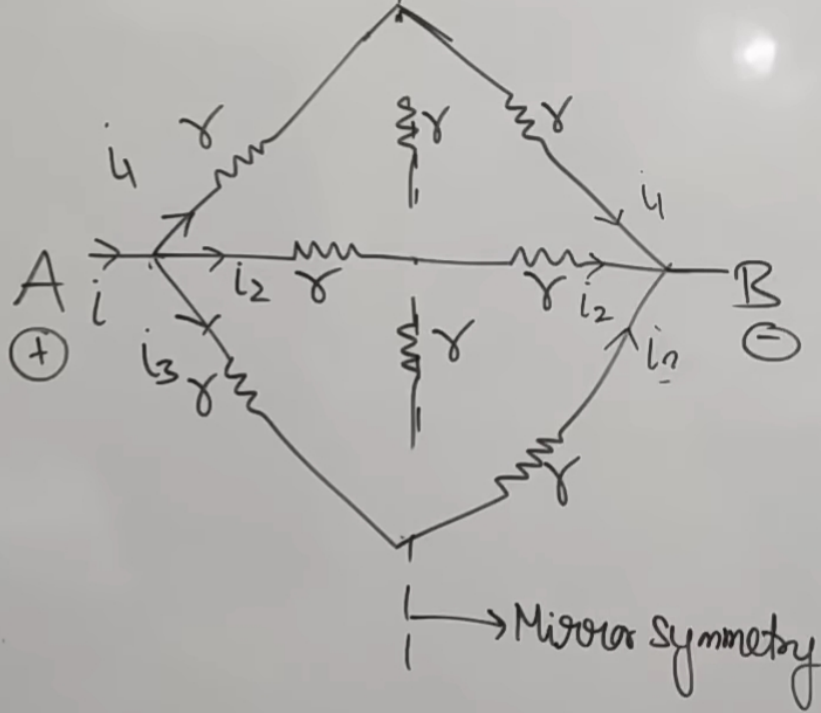
Mirror Symmetry:

②



Maxwell's Symmetry:

②

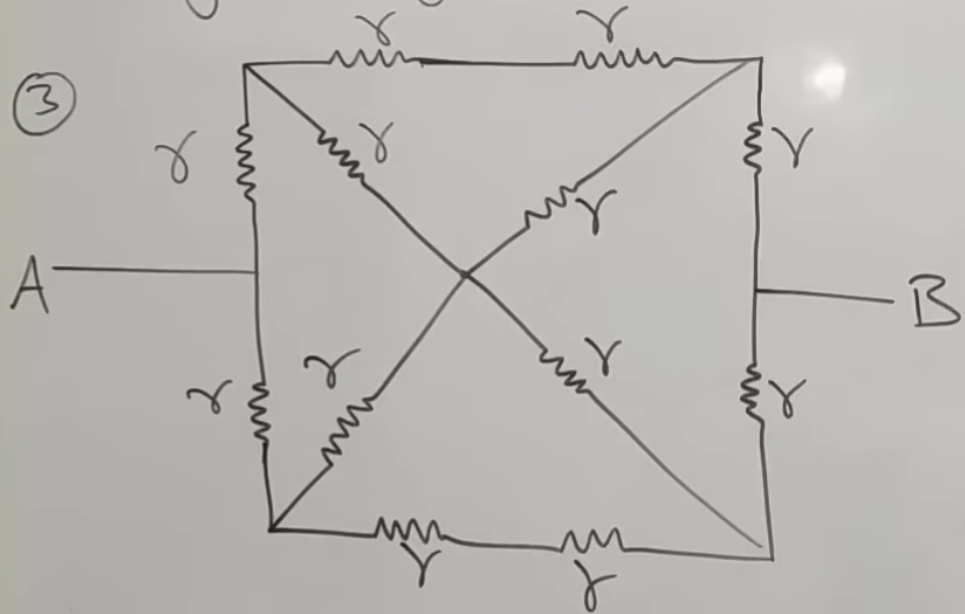


$$R_e = \frac{R}{n}$$
$$= \frac{2r}{3}$$

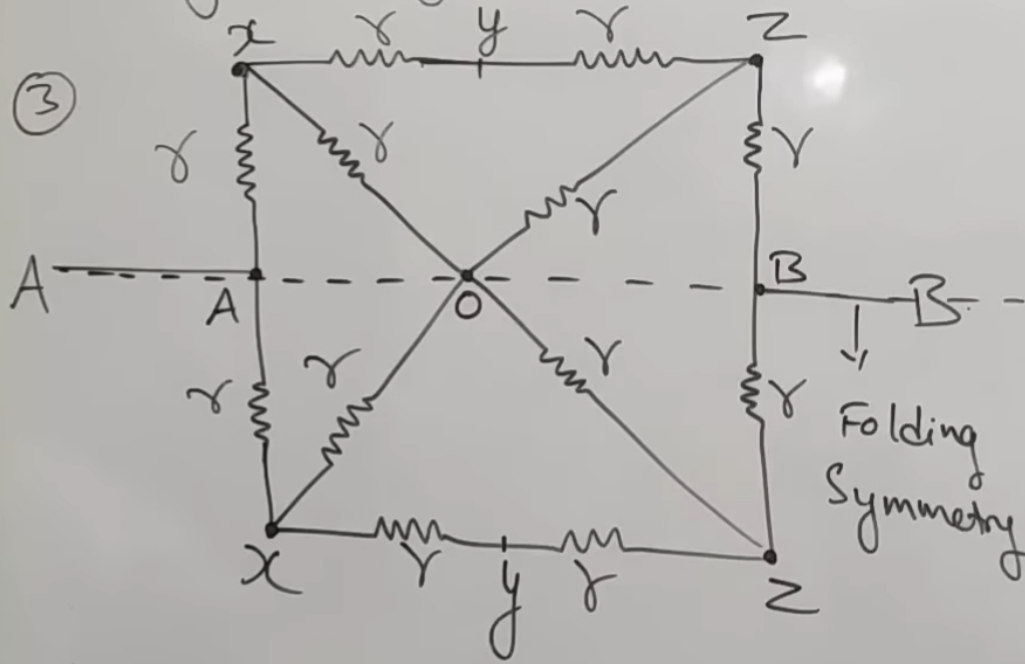
$$\frac{1}{R_e} = \frac{1}{2r} + \frac{1}{2r} + \frac{1}{2r} = \frac{3}{2r}$$

$$R_e = \frac{2r}{3}$$

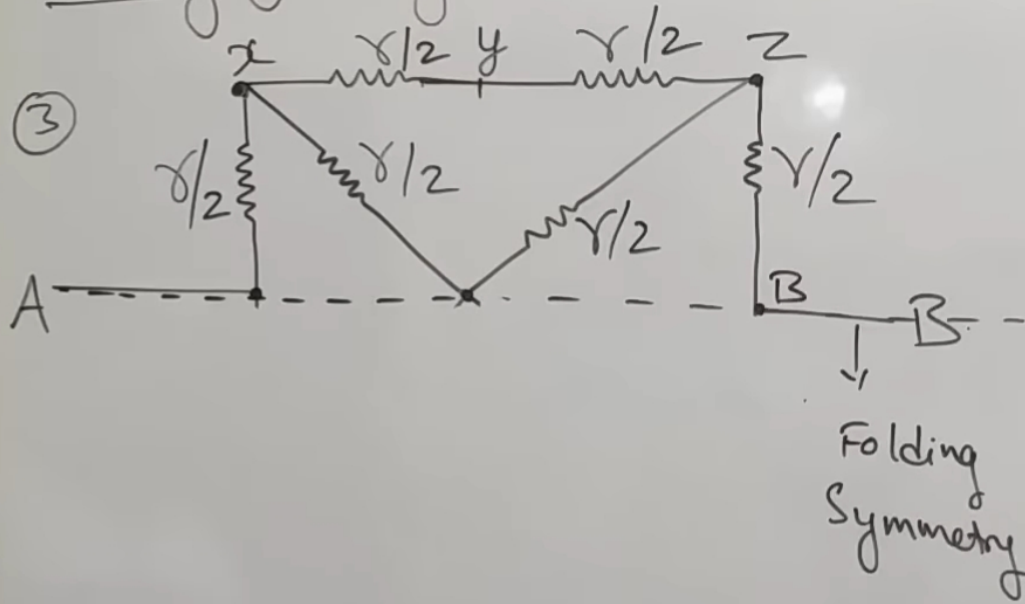
Folding Symmetry.



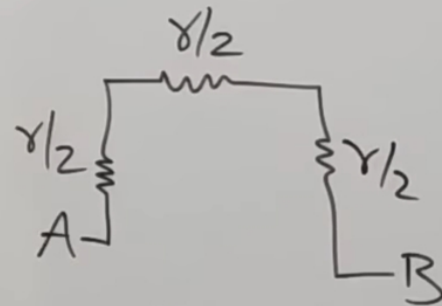
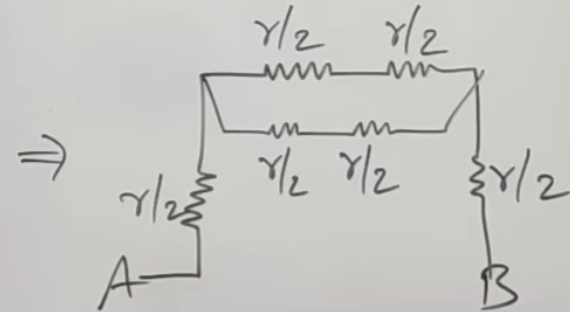
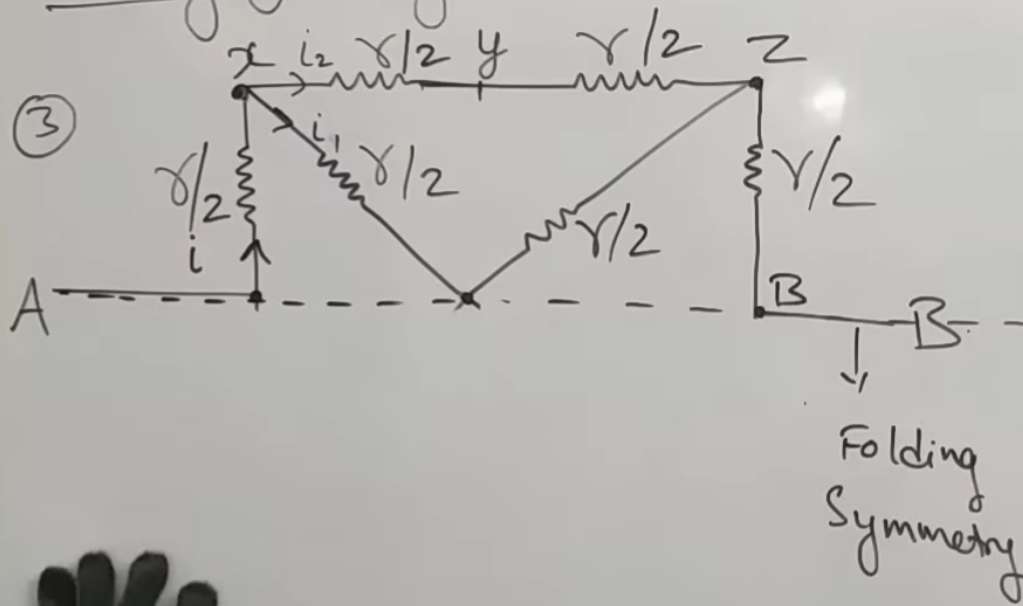
Folding Symmetry.



Folding Symmetry.



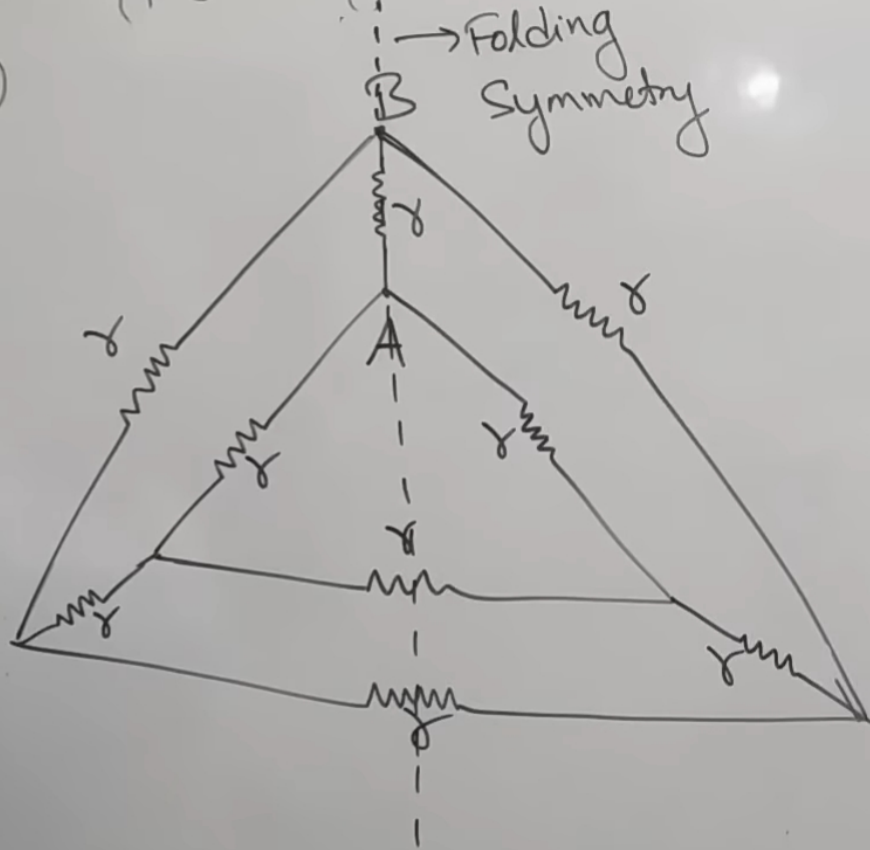
Folding Symmetry.



$$R_e = \frac{3r}{2}$$

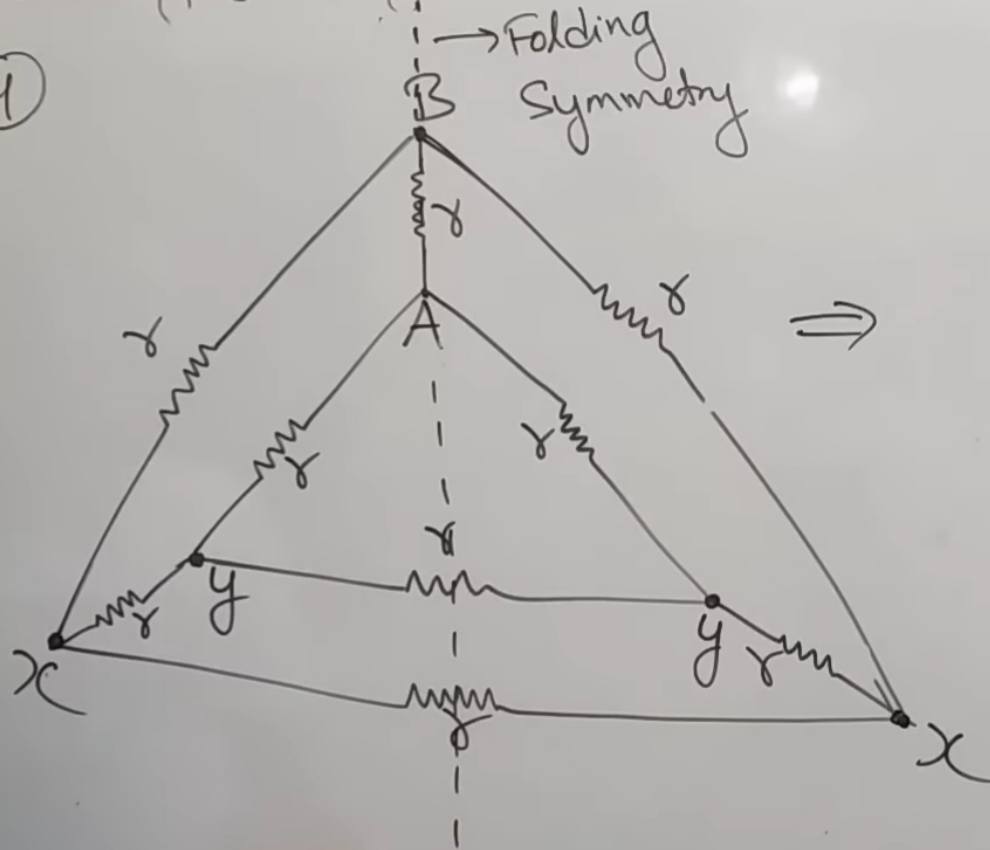
Folding Symmetry

(4)

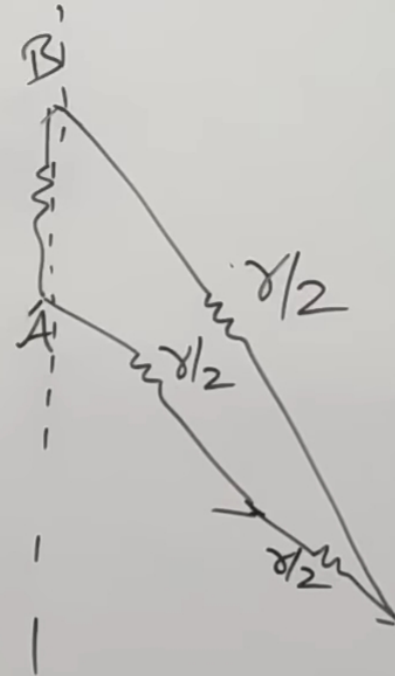


Folding Symmetry

(4)

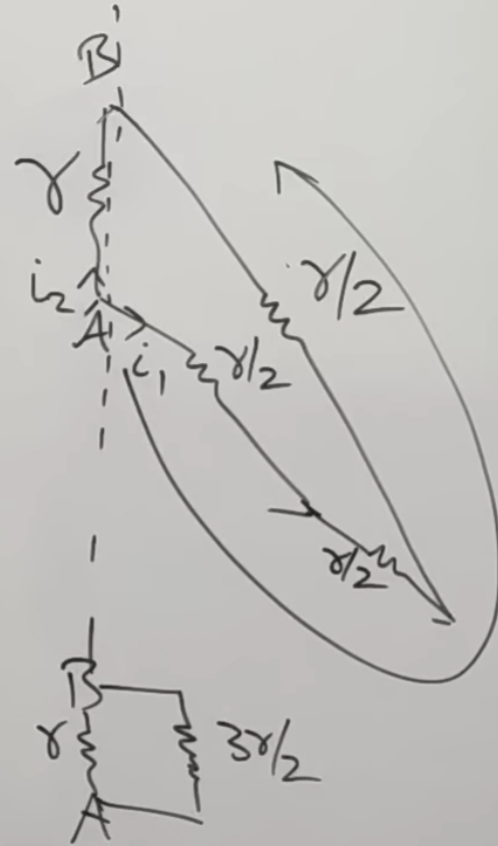
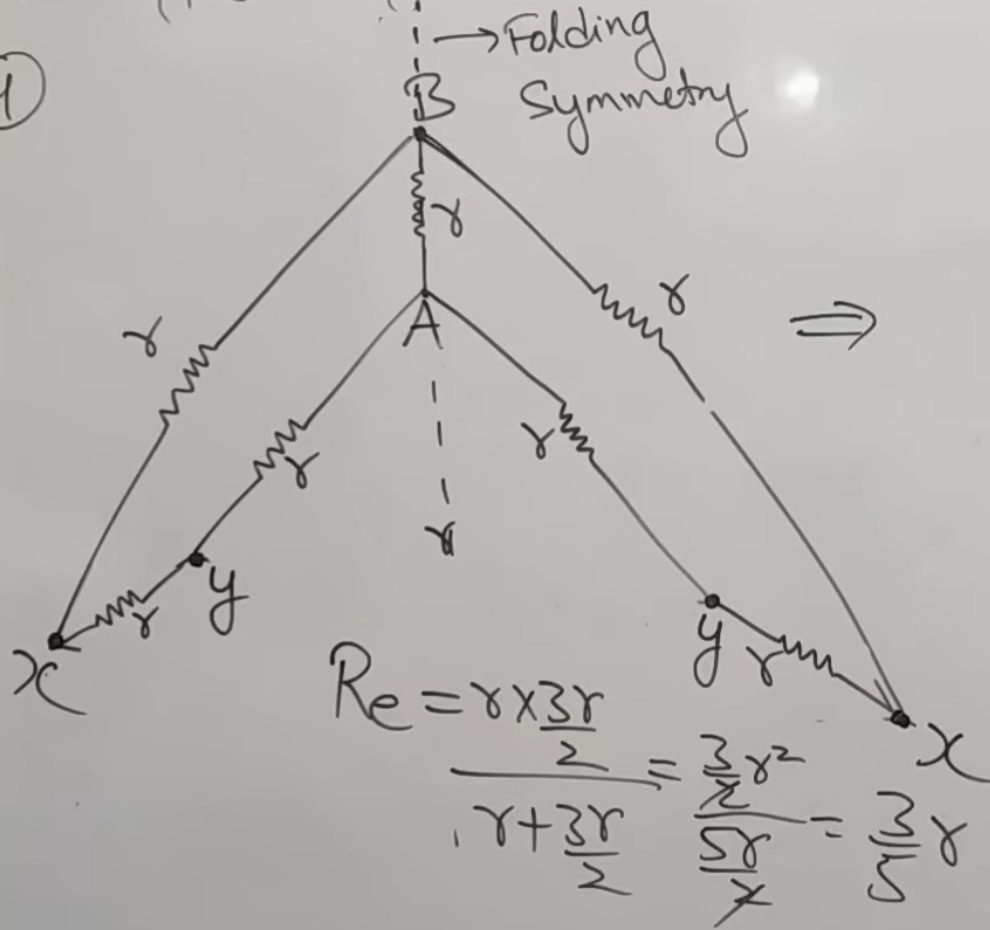


$\Rightarrow$

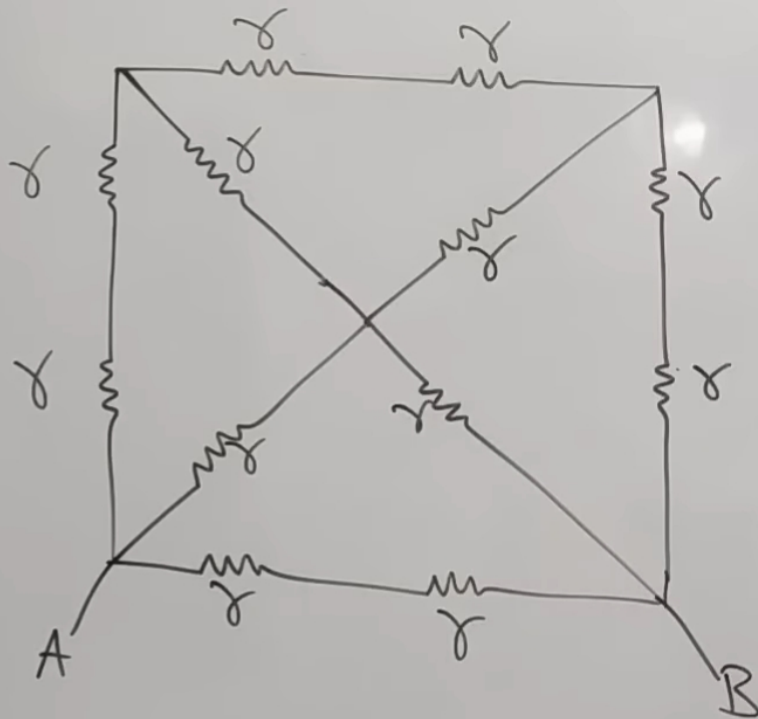


Folding Symmetry

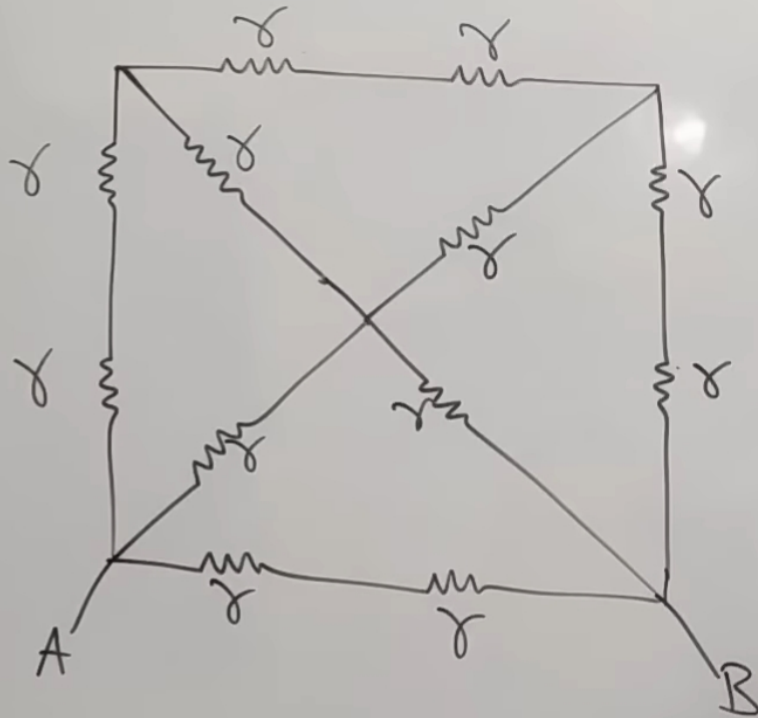
(4)



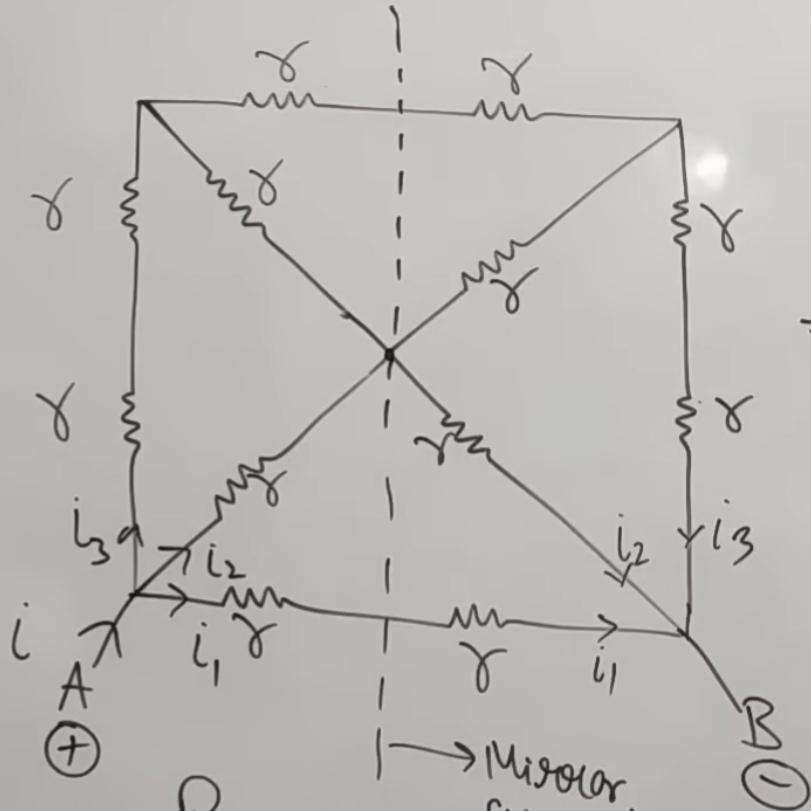
5



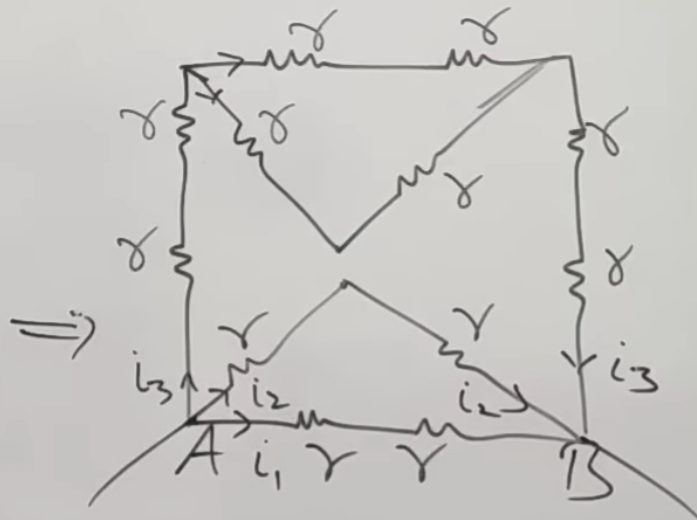
5



5



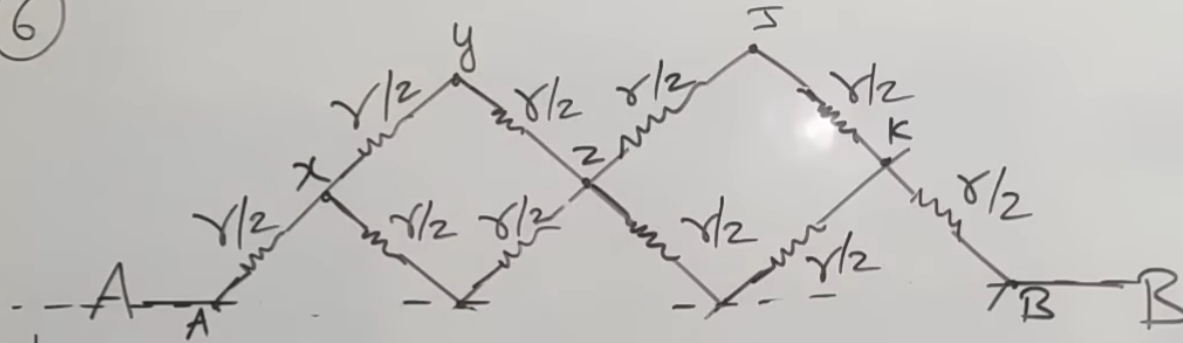
$$R_e = \frac{5r \cdot r}{5r + r} = \frac{5}{6}r$$



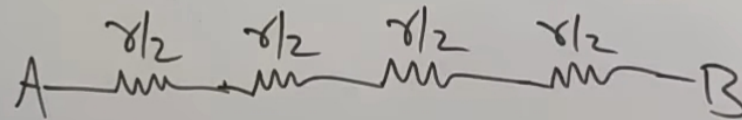
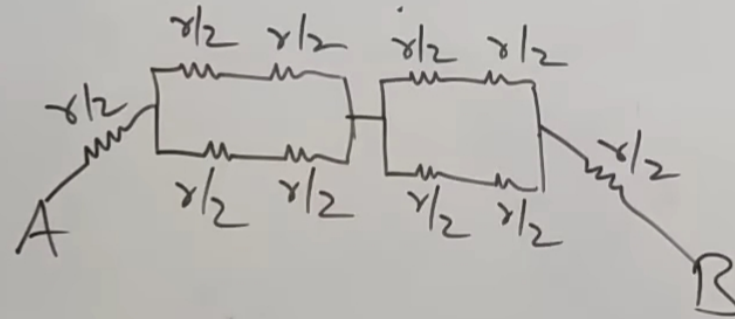
⑥



⑥



↓
Folding
Symmetry



$$R_e = 2r.$$